Applications of Information and Communication Technologies

Lecture-2

Today's Agenda

- Introducing today's technologies
- History of Computers,
- Classification of computers
- Data and Information

Today's Technology

- Technology makes us able to work more affectively and efficiently:

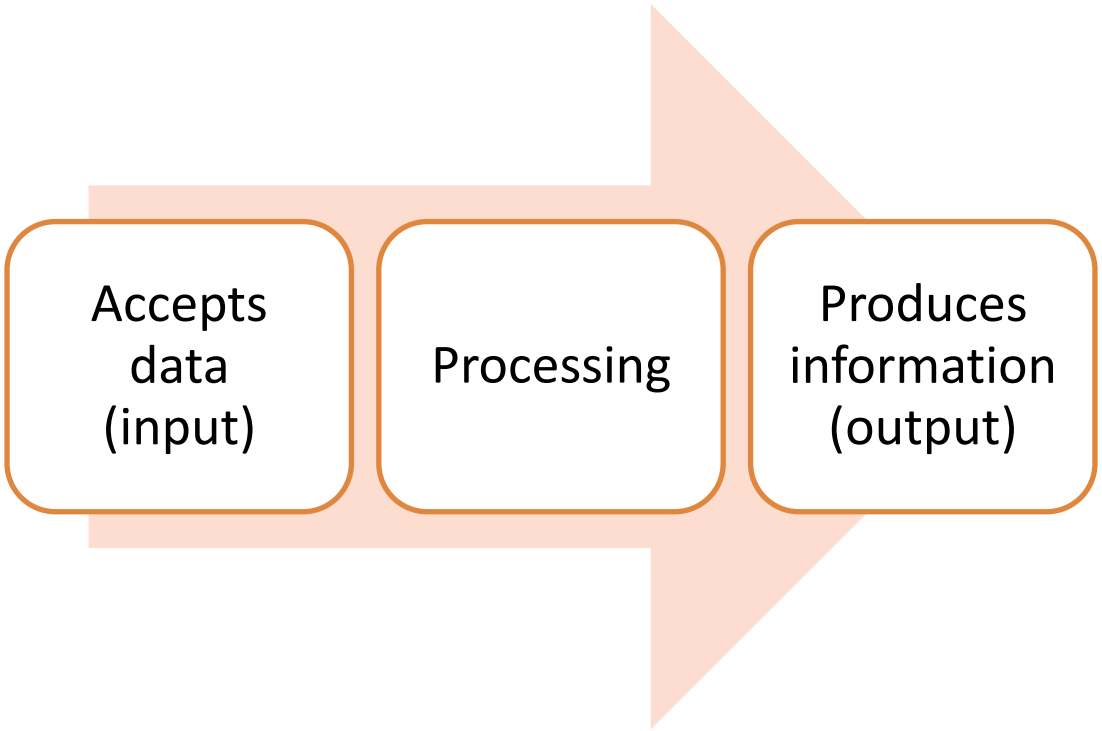
Today's Technology

- Technology makes us able to work more affectively and efficiently:
  - Access and search info
  - Share personal ideas, photos, video
  - Communicate with others
  - Manage finances
  - Shop for goods and services
  - Play games
  - Entertainment
  - Business activities
  - Keep your life and activities organized

Digital Literacy

- Digital literacy involves having a current knowledge and understanding of computers, mobile devices, the web, and related technologies.

What is Computer?



What is Computer?

- Derived from word “compute” means to calculate

- Computer is a modern electronic machine that is used to **solve different kinds of problems** according to the set of instruction given to it.

What is Computer?

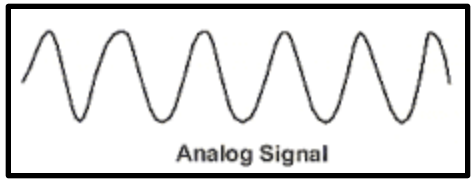
- A computer is an electronic machine that operates under the control of instructions, that
  - accepts the **input data**,
  - process the given data according to the predefined **instructions**,
  - produces **output data** and
  - **stores** the result for future use.
  - It can also **retrieve** the stored results as and when required again

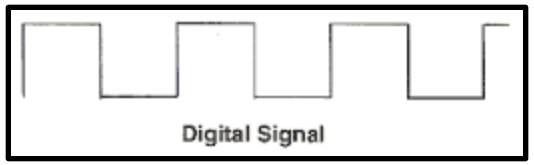
Fundamental Characteristics of Computer

- Speed
- Accuracy
- Reliability
- Storage
- Retrieving Data & Programs
- Automation
- Communications

Classification of Computers

- Operating Mechanism
  - Analog → data as continuous electronic signals
  - Digital → data as binary number i.e. 0 or 1
  - Hybrid → Analog + Digital e.g. ECG, Heartbeat computers





- Use
  - General Purpose and Special Purpose
- Size
  - Micro, Mini, Mainframe, Super

Personal Computer: PC

- The six primary types of computers in this category are
- Desktop computers
- Workstations
- Notebook computers (Laptops)
- Tablet computers
- Handheld computers
- Smart phones

Laptop

Laptops

A **laptop**, also called a *notebook computer*, is a thin, lightweight mobile computer with a screen in its lid and a keyboard in its base (Figure 1-2). Designed to fit on your lap and for easy transport, most laptops weigh up to 7 pounds (varying by manufacturer and specifications). A laptop that is less than one inch thick and weighs about three pounds or less sometimes is referred to as an ultrathin laptop. Most laptops can operate on batteries or a power supply or both.



Figure 1-2 A typical laptop has a keyboard in the base and a screen in the lid, with the lid attaching to the base with hinges.

Tablet

Tablets

Usually smaller than a laptop but larger than a phone, a **tablet** is a thin, lighter-weight mobile computer that has a touch screen (read How To 1-1 for ways to interact with a touch screen). A popular style of tablet is the slate, which does not contain a physical keyboard (Figure 1-3). Like laptops, tablets run on batteries or a power supply or both; however, batteries in a tablet typically last longer than those in laptops.



Figure 1-3 A slate tablet.

© iStockphoto / franckreporter

Desktops and All-in-Ones

A **desktop**, or desktop computer, is a personal computer designed to be in a stationary location, where all of its components fit on or under a desk or table. On many desktops, the screen is housed in a display device (or simply display) that is separate from a tower, which is a case that contains the processing circuitry (Figure 1-4a). Another type of desktop called an **all-in-one** does not contain a tower and instead uses the same case to house the display and the processing circuitry (Figure 1-4b). Some desktops and all-in-ones have displays that support touch.

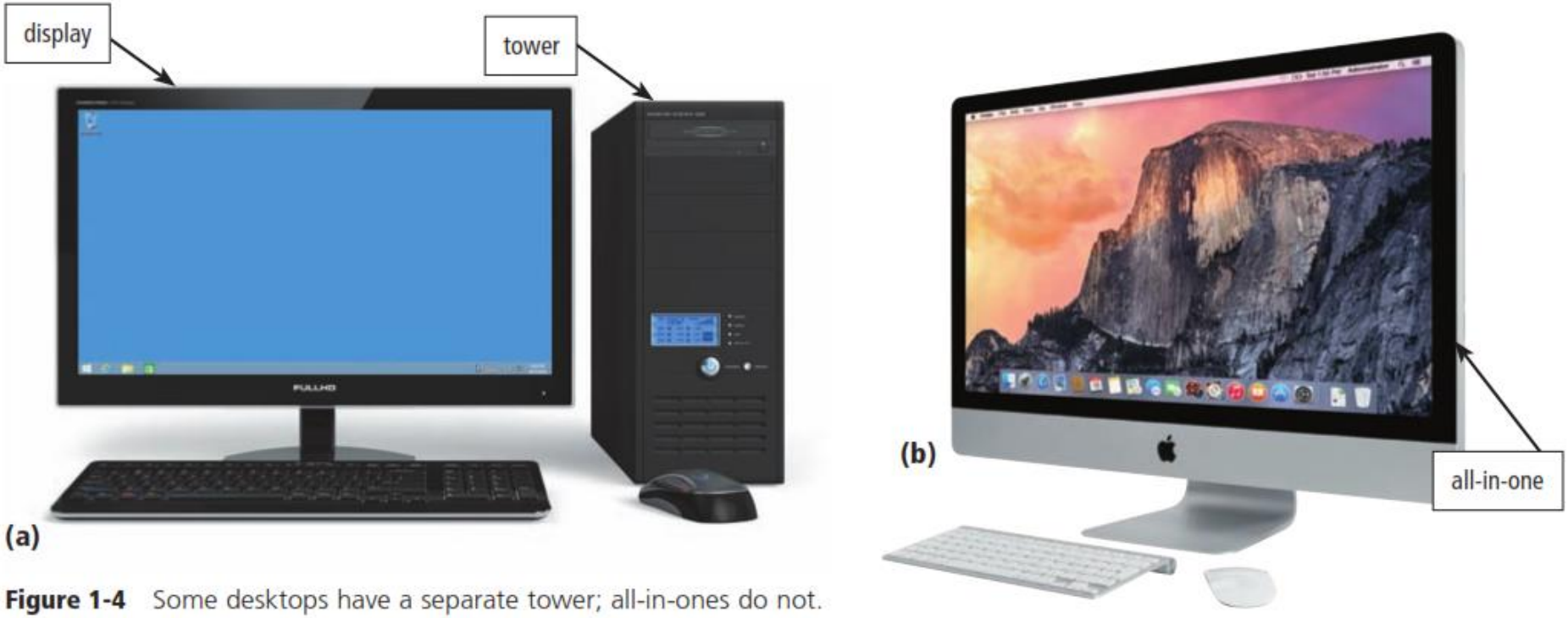


Figure 1-4 Some desktops have a separate tower; all-in-ones do not.

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Computers for Organizations

- Network Servers
  - A powerful PC with special software and equipments
- Mainframe Computers
  - Large organizations with millions of records a day, handle large data bases
  - Large size
- Mini Computers
- Super Computers
  - Fastest
  - Complex operations at a very high speed

Computers For Organizations

Network servers

- Centralized computer
- All other computers connect
- Provides access to network resources
- Multiple servers are called **server farms**
- Often simply a powerful desktop



Figure 1-5 A server provides services to other computers or devices on a network.

Computers For Organizations

Mainframes

- Used in large organizations
- Handle thousands of users
- Users access through a terminal



Computers For Organizations

- **Minicomputers**
  - Called midrange computers
  - Power between mainframe and desktop
  - Handle hundreds of users
  - Used in smaller organizations
  - Users access through a terminal

Computers For Organizations

- **Supercomputers**
  - The most powerful computers made
  - Handle large and complex calculations
  - Process trillions of operations per second
  - Found in research organizations



Computers For Organizations

- Pakistan's most powerful supercomputers:
  - Pak-Austria Fachhochschule, Haripur.
  - SINES, NUST, Islamabad
  - HEC
  - PIEAS, Islamabad
  - LUMS in Lahore.
  - NED in Karachi.
  - Khan Research Laboratories (KRL) and the Pakistan Institute of Nuclear Science & Technology, since the 1990s.

Mobile and Game Devices



Smartphone



Digital camera



Portable and digital media player



E-book reader



Wearable device

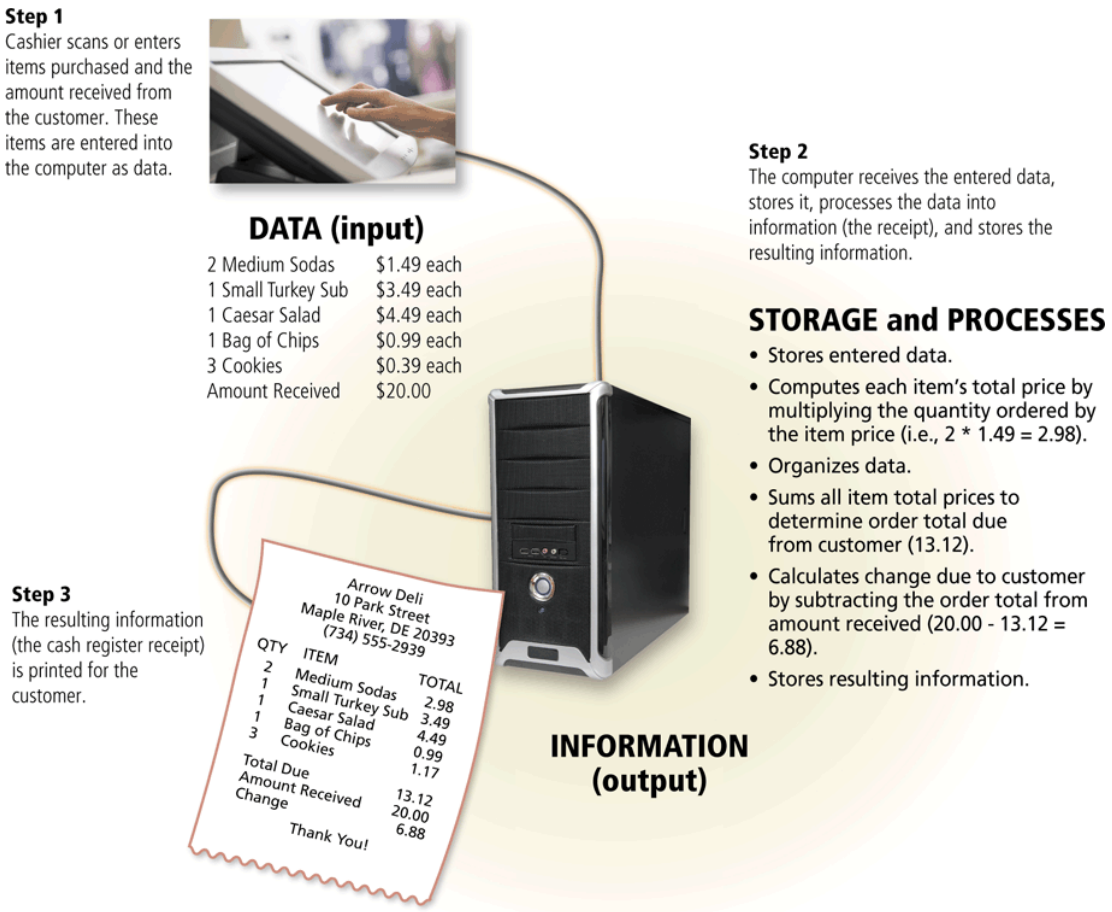


Game device

Data & Information

- **Data** refers to the facts concerning things such as people, object
  - Data is defined as collection of **raw facts** and figures, which is collected for specific purpose
- Processed and refined data is referred to as **information**
  - When raw facts and figures are **arranged** in such a suitable manner that they give the clear and proper **meaning**

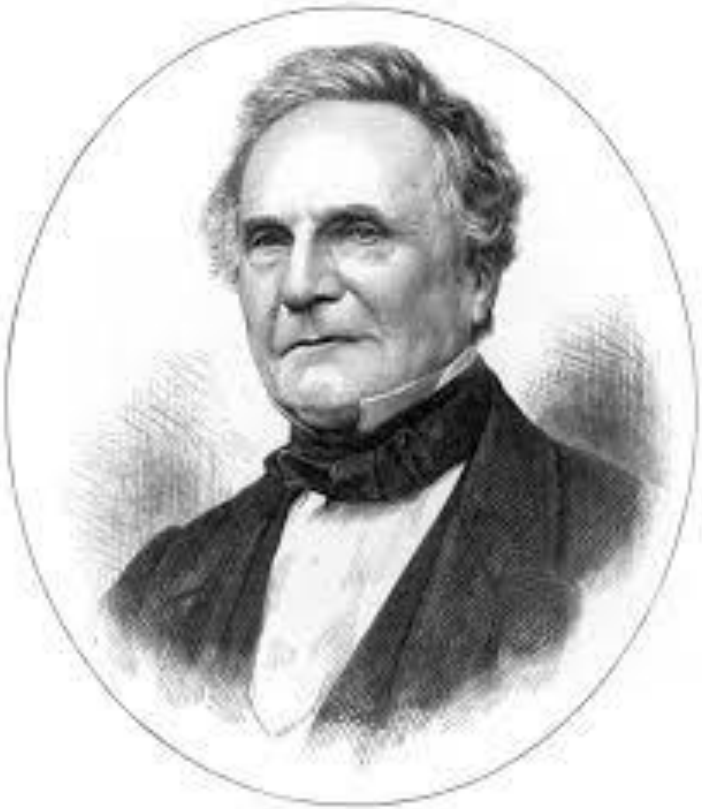
Data and Information



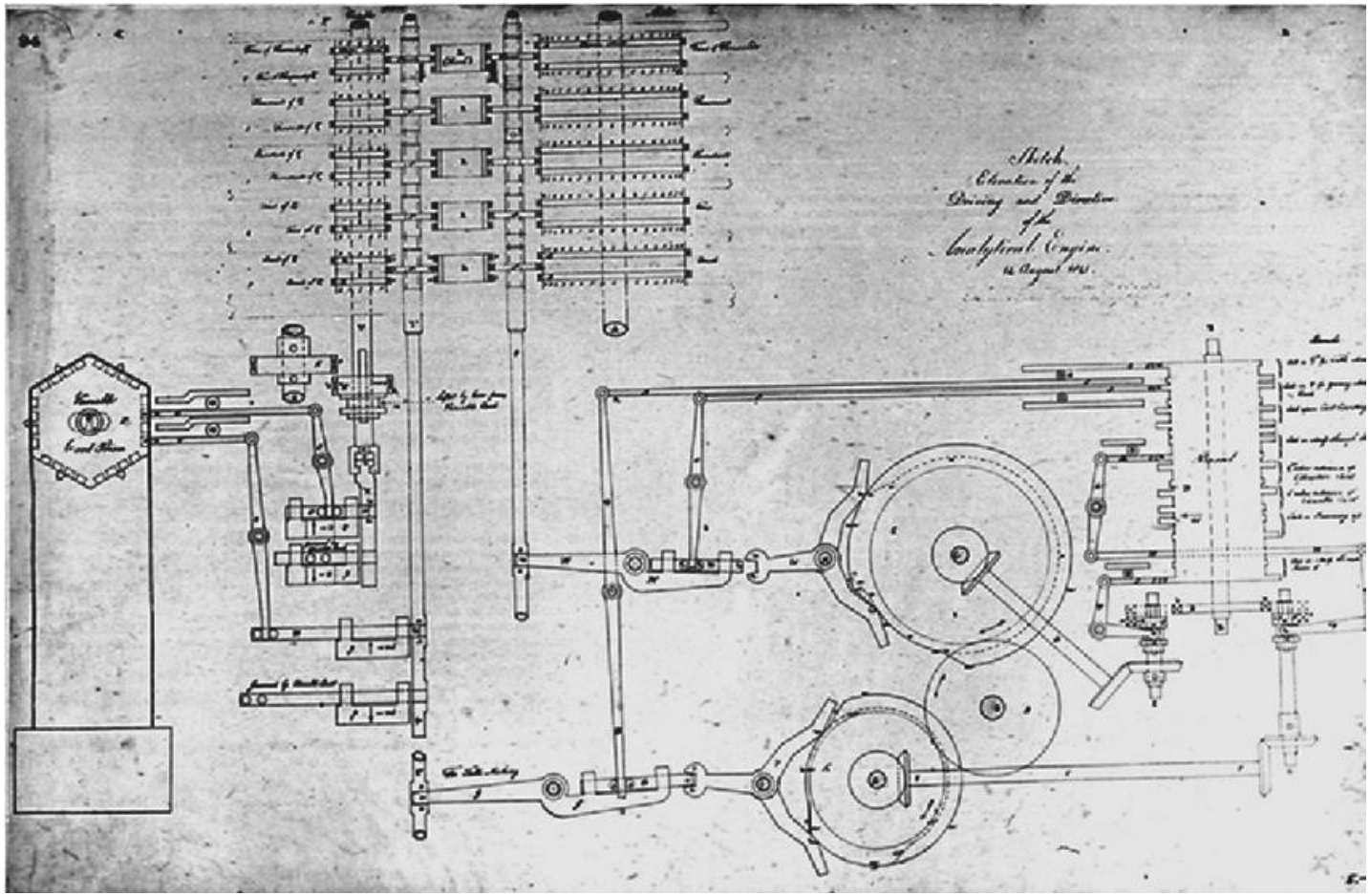
HISTORY OF COMPUTERS

Charles Babbage

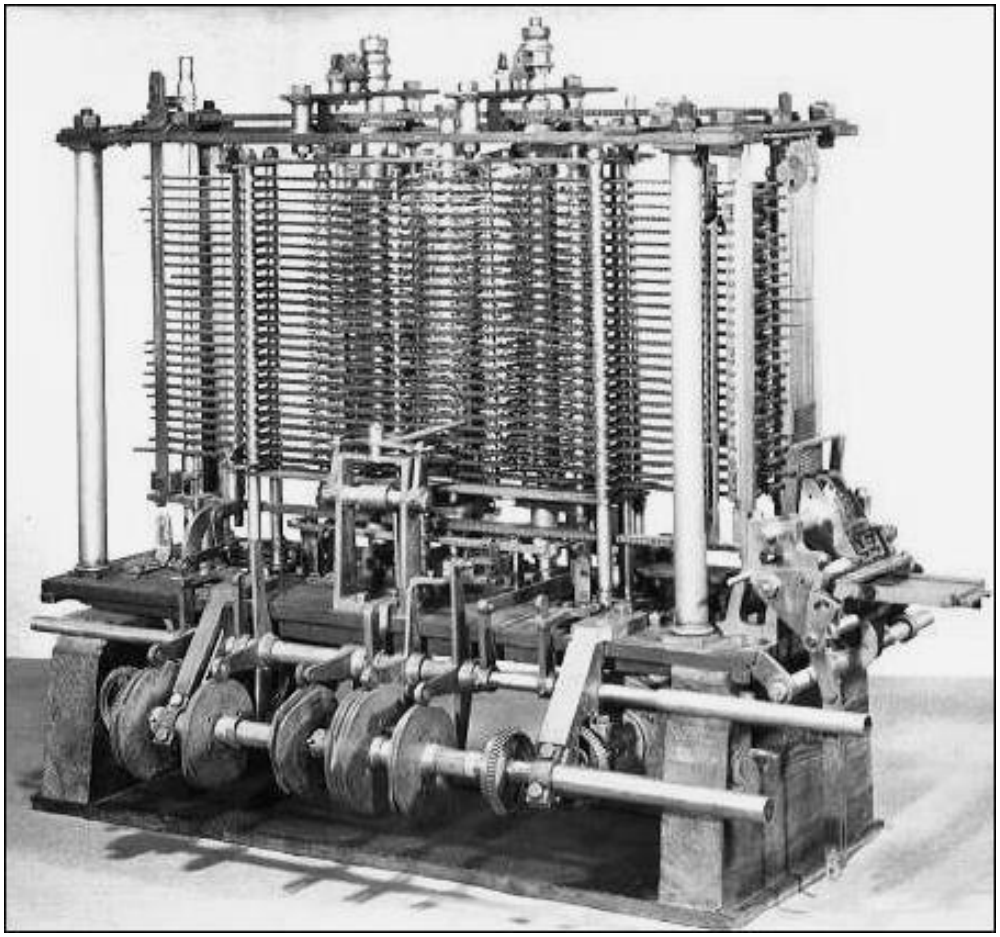
- Nineteenth century
- British mathematician
- Designed “analytical engine”
- His design was too complex for him to build with the technology of his day, so it was never implemented.
- Included a memory so that intermediate values did not have to be re-entered.



Analytical Engine



Analytical Engine



Ada Augusta

- Ada, the daughter of Lord Byron (the English poet)
- A skilled mathematician.
- First programmer.
- When she was 18, Ada visited the Mechanics Institute to hear lectures on the “Difference Engine”, built by Charles Babbage. It was said that, upon seeing Babbage’s machine, Ada was the only person in the room to understand immediately how it worked and to recognize its significance.

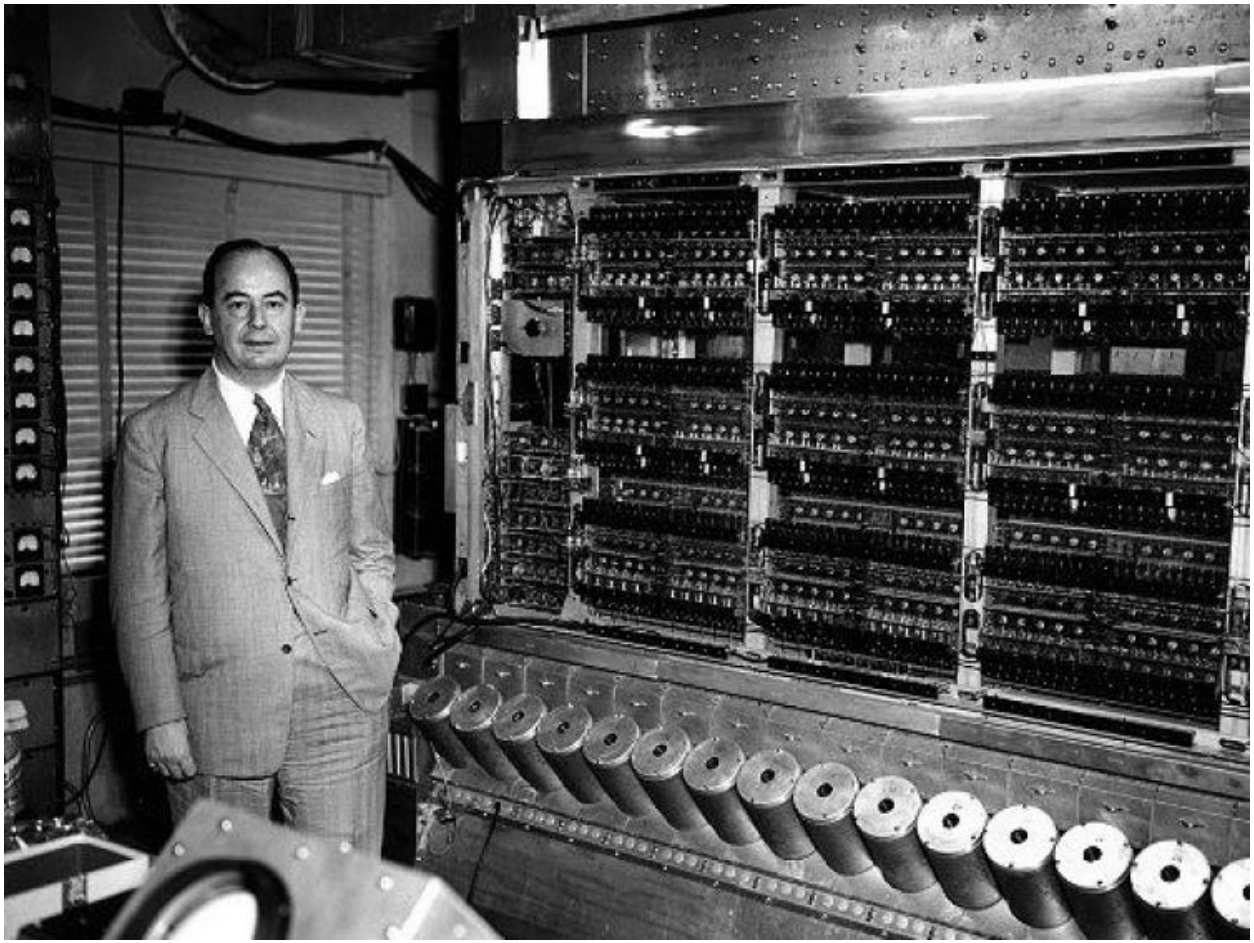


Alan Turing

- British mathematician
- invented an abstract mathematical model called a Turing machine,
- The most prestigious award given in Computer Science is the Turing Award, named for Alan Turing.



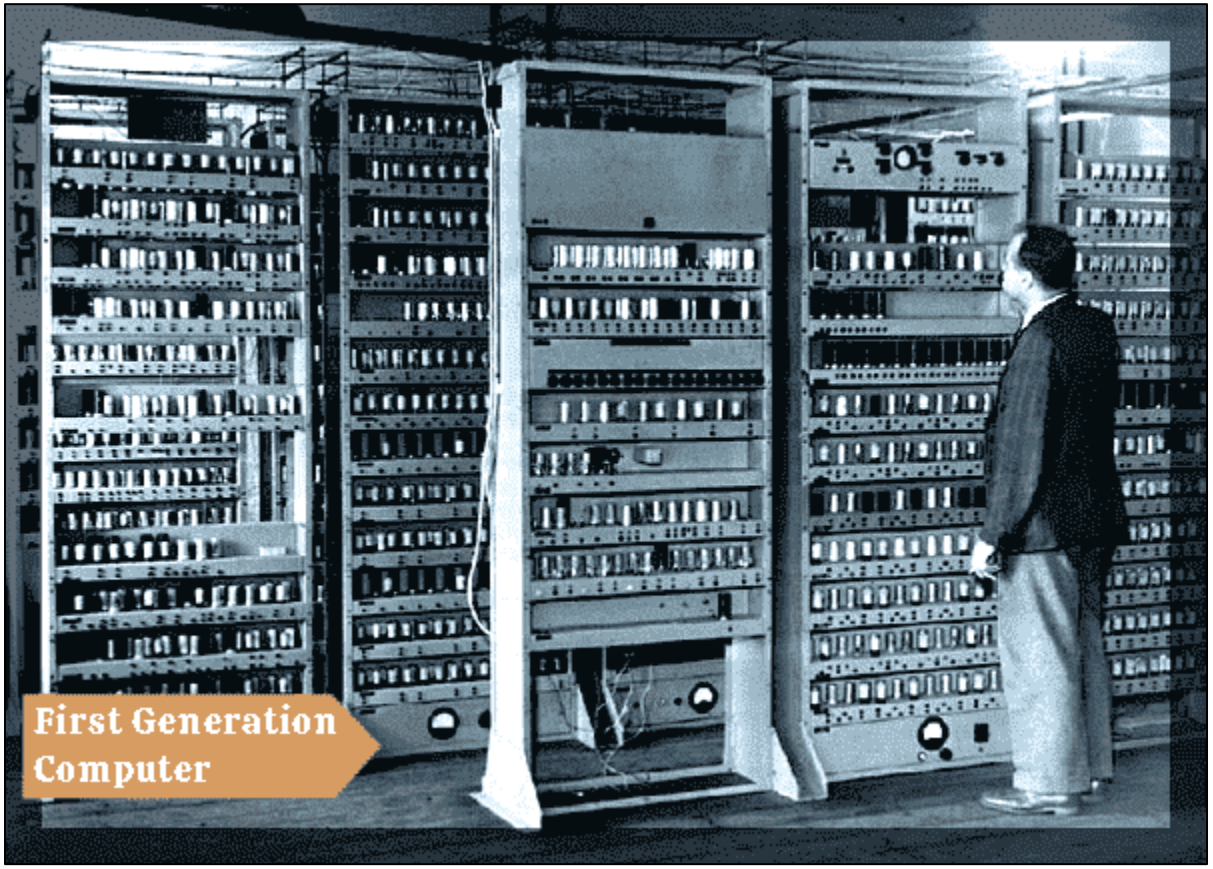
Turing Machine



World War II period

- By the outbreak of World War II, several computers were under design and construction. The Harvard Mark I and the ENIAC are two of the more famous machines of the era.
- **John von Neumann**, who had been a consultant on the ENIAC project, started work on another machine known as EDVAC, which was completed in 1950.
- In 1951 the first commercial computer, UNIVAC I, was delivered to the Bureau of the Census. The UNIVAC I was the first computer used to **predict the outcome of a presidential election**. The early history that began with the **abacus** ended with the delivery of the UNIVAC I. With the delivery of that machine, the dream of a device that could rapidly manipulate numbers was realized; the search was ended.

ENIAC



Fast Forward

- Some experts predicted at that time that a small number of computers would be able to handle the computational needs of mankind. What they didn't realize was that the ability to perform fast calculations on large amounts of data would radically change the very nature of fields such as mathematics, physics, engineering, and economics.
- After 1951 the story becomes one of the ever-expanding use of computers to solve problems in all areas. From that point, the search has focused not only on building faster, bigger devices, but also on the development of tools to allow us to use these devices more productively. The history of computing hardware from this point on is categorized into several generations based on the technology they employed.

5 Generations of Computers

- First Generation of Computers (1940-1956)
  - Vacuum Tubes
  - Input: Punch card, Output: Print out
  - Able to handle complex problems
  - Example: ENIAC, EDVAC, EDSAC, UNIVAC
  - Limitations:
    - Machine Language
    - Single tasking
    - Special Purpose → Inflexible
    - Large and bulky
    - Temperature maintenance

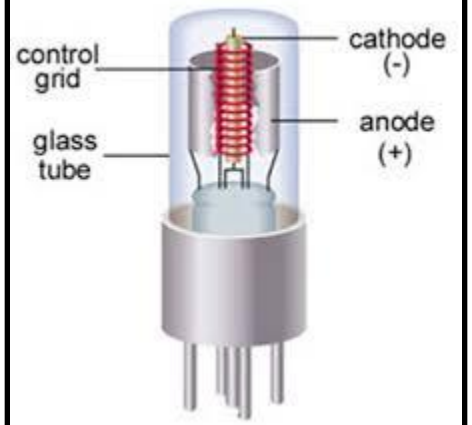
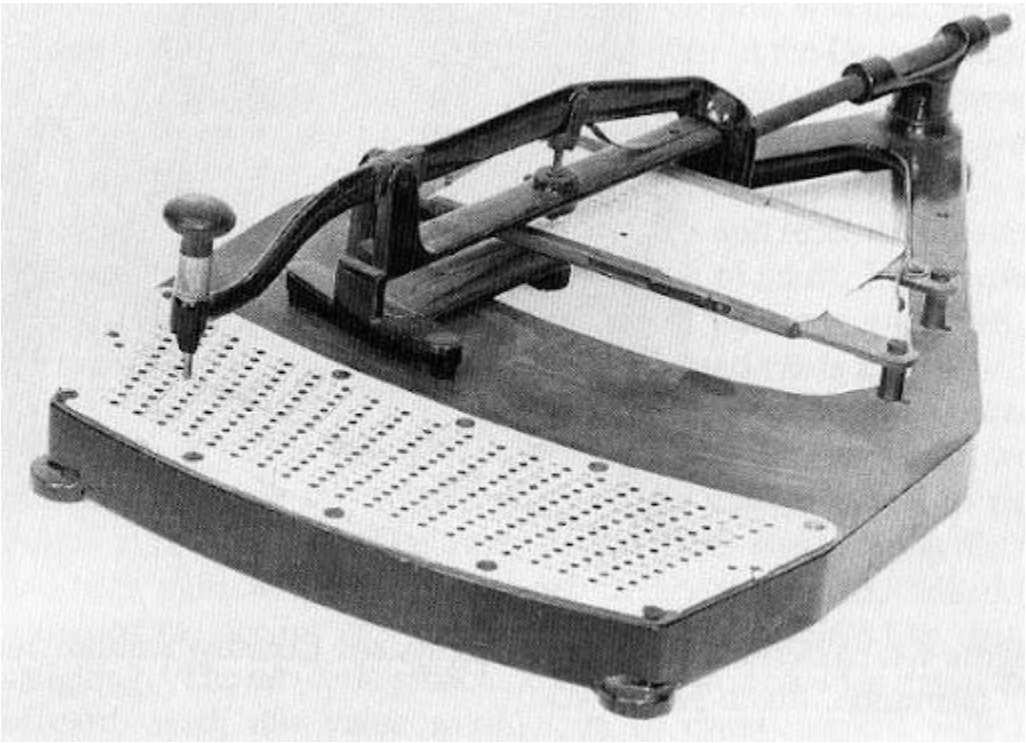


Image: Vacuum Tube

Punch Card



5 Generations of Computers (contd.)

- Second Generation of Computers (1956-1963)
  - Transistors
  - Invention of Printers, Secondary storage, Operating systems
  - Input: Punch card, Output: Printout
  - Assembly Language
  - Smaller size
  - Limitations:
    - Very costly, unaffordable to home users
    - Special purpose → Inflexible
    - Required air-conditioned places

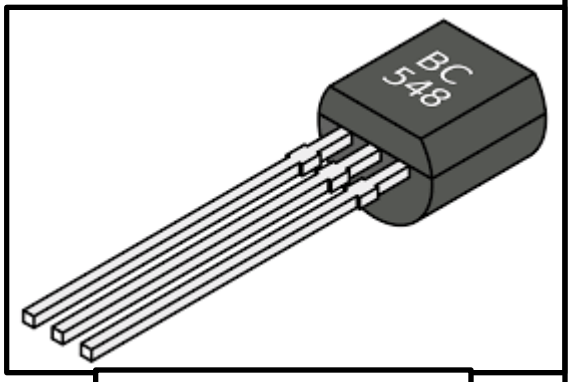


Image: Transistor

5 Generations of Computers (contd.)

- Third Generation of Computers (1964-1975)
  - Integrated Circuits (ICs)
  - Small computational time (within nanoseconds)
  - Easily portable
  - High-Level Languages
  - General-purpose Computers
  - More reliable than previous computers
  - Limitations:
    - Small storage capacity
    - High cost
    - Required air-conditioned places



Image: Integrated Circuit

5 Generations of Computers (contd.)

- Fourth Generation of Computers (1975-1989)
  - Large Scale Integration (LSI) and Very Large Scale Integration (VLSI)
  - Thousands of transistors fitted onto one silicon chip
  - Invention of microprocessor
  - Invention of Personal Computers
  - Use of RAM
  - Small, cheap, compact, powerful
  - Development of Graphical User Interface (GUI)
  - Development of Local Area Network (LAN)
  - Limitations:
    - Dependent on programmer for instructions





5 Generations of Computers (contd.)

- Fifth Generation of Computers
  - Ultra Large Scale Integration (ULSI)
  - 10million components on one small chip
  - Development of laptops,
  - Development of Parallel Processors
  - Development of servers
  - Development of internet
  - Limitation
    - Not intelligent



References

- Discovering Computers
  - Chapter-1